

CSCI212 : Lecture #10

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Review

- Sub-class/Super-class relationship
- **extends** Java keyword to do this:

```
1      class B extends A { //Makes B as a sub-class, and A as a super-class}
2
```

Code example #1 (see the Java files on GitHub)

Let us consider the following system:

Employee

```
1      Employee
2          name
3          getName()
4          weeklyPay()
5          toString()
```

HourlyEmployee (inherits from Employee)

```
1      hoursWorked
2      hourlyRate
3      weeklyPay() // An 'overwritten' method
```

SalariedEmployee (inherits from Employee)

```
1      annualSalary
2      weeklyPay() // An 'overwritten' method
```

In this example, **Employee** is a super-class of **HourlyEmployee** and **SalariedEmployee**.

For-each loop

A *for-each* loop will iterate through the contents of a container.

```
1      // The general syntax is
2      for(ObjectType iterator : ContainerType) {
3          // for loop contents
4      }
```

Dynamic Dispatch

the method chosen is determined by the stored object's type at **run time**. View lines 17-34 of **SalariedEmployee.java** in Ryba.lecture10 notes on GitHub.